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Original Article

Formation of laminated nano-coatings for enhanced anti-oxidation and electromagnetic wave absorbing properties of carbon fiber

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ABSTRACT

In the paper, a simple and reliable method was proposed. A novel metal/ceramic laminated nano-coating (Ni-P/SiC) was prepared on the surface of carbon fiber (CF) by plasma electrolytic spraying and electroless plating. The morphologies, composition, antioxidation, and electromagnetic wave absorbability of the laminated nano-coatings have been studied. The thickness of the coating was 392.8nm measured by scanning electron microscopy (SEM). The coated carbon fiber not only has good resistance to high temperature oxidation, but also has high electromagnetic absorption in the range of 2∼18 GHz. The minimum reflection loss (RL) of the laminated Ni-P/SiC coated CF decreased from −3.9 to −26.95 dB, indicating that 99% of the electromagnetic waves were absorbed. The weight loss rate was reduced and the complete weight loss temperature was increased from 950 to 1050°C. The electromagnetic wave absorption mechanism was discussed in detail.

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1. Introduction

At present, electromagnetic interference (EMI) problems greatly affect people's daily lives and disrupt commercial and industrial equipment. Therefore, the method of preventing electromagnetic interference has always been an important issue for civil and military purposes [1–6]. In order to solve the problem of electromagnetic interference, electromagnetic absorption materials have received more and more attention in recent years. In addition, it has broad applications in radar systems, civil and military fields [7–10].

The new absorbing materials are required to be thin, light weight, strong absorption, absorption frequency bandwidth, multi-band compatible stealth and other comprehensive performance indicators. Carbon fiber has been today on the central stage of structural absorbing materials due to its light weight, high strength and good heat resistance [11–13]. The study of structural absorbing materials also bid farewell to the old era when metal was used as the main bearing component because of the emergence of high performance carbon fibers. However, since carbon fiber has a low electrical resistivity and almost totally reflects the incident wave, only the

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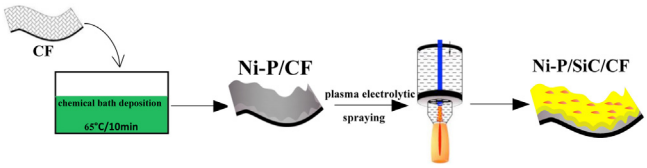

Fig. 1 – The schematic process diagram of preparing laminated nano-coating on carbon fibers.

modified carbon fiber material can be used as an absorbing material. The research of metal absorbing materials mainly focuses on simple metals or alloys of magnetic metals such as Fe, Co and Ni. Magnetic absorbing properties of the magnetic materials are associated with high dielectric loss and magnetic loss. Common inorganic absorbing materials include SiC, SiN and $BaTiO_3$. SiC is the main component of multi-band absorbing materials, and its loss mechanism is mainly based on dielectric loss. It is a typical high temperature absorbing material [14–20]. Zhou et al. [21] prepared SiC coating on carbon fiber surface by CVD with PyC and BN interphases to compare their dielectric and microwave absorption properties. The results showed that the minimal RL of Cf-PyC-SiC is achieved at −10.5 dB, and Cf-BN-SiC with a minimal RL of −13.3 dB. However, CVD is limited by its low deposition rate, high cost and high reaction temperatures. Huang et al. [22] studied on microwave absorbing properties of FeCo-coated reinforced carbon nanofibers. The optimized reflection loss value of −24.05 dB was found at 14.88 GHz. Zhou et al. [23] studied the *in-situ* growth of silicon carbide nanofibers on the surface of carbon fibers by catalytic chemical vapor deposition. When the thickness of the specimen is 2 mm, the lowest value of the reflectivity can approach −19.9 dB. Meanwhile, carbon fiber is rapidly oxidized when the temperature exceeds 400 °C in aerobic environment, which limits the application of carbon fiber. A good way to overcome these limitations is to effectively prevent them from coming into contact with oxygen. Applying SiC coating on carbon fiber can solve this problem [24–26]. However, it is difficult to prepare SiC coating on the surface of carbon fiber quickly and easily.

In this paper, the Ni-P coating was deposited on the surface of carbon fiber by electroless plating, and then the SiC coating was prepared by plasma electrolytic spraying. As the underlayer of the laminated coating, Ni can effectively protect the carbon fiber from damage. In the laminated coating, Ni was used as a magnetic loss material, and SiC was used as a dielectric loss material. The laminated coating has superior absorbing properties for carbon fiber. The surface morphology, thermal stability and electromagnetic properties of the coated carbon fiber were discussed in detail. The results show that the coating not only has good absorbing properties, but is also an excellent antioxidant.

2. Experimental

2.1. Materials

In the experiment, using commercialized T300 carbon fiber of Xiangsheng Carbon Fiber Technology Co., Ltd. All the chemicals and reagents are analytical grade. Acetone (CH_3COCH_3), nitric acid (HNO_3), tin dichloride ($SnCl_2{\cdot}2H_2O$), hydrochloric acid (HCl), palladium chloride ($PdCl_2$), nickel salt ($NiSO_4{\cdot}6H_2O$), Sodium hypophosphite ($NaH_2PO_2{\cdot}H_2O$), ammonia chloride (NH_4Cl), sodium citrate ($Na_3C_6H_5O_7{\cdot}2H_2O$), ammoniawater ($NH_3{\cdot}H_2O$) were purchased from Beijing Tongguang Fine Chemical Co., Ltd.

Table 1 – The chemical components and concentration in the plating bath.

Component	Concentration (g/L)
$NiSO_4{\cdot}6H_2O$	30
$NaH_2PO_2{\cdot}H_2O$	20
NH_4Cl	20
$Na_3C_6H_5O_7{\cdot}2H_2O$	25

2.2. Preparation of laminated nano-coatings

Fig. 1 shows the schematic process diagram of preparing laminated nano-coating on carbon fibers. Ni-P coating was initially prepared on carbon fiber surface by the electroless deposition method. Firstly, carbon fiber was soaked in the acetone solution at room temperature (RT) for 30 min. Then the carbon fiber was washed in the distilled water several times to remove all residues on the fibers. Secondly, carbon fiber was coarsened in 65% nitric acid solution for 40 min. Thirdly, fibers was sensitized in 20 g/L $SnCl_2$ and 3.4 mol HCl solution for 15 min. Fourthly, fibers was activated in 0.25 g/L $PdCl_2$ and 0.03 mol HCl solution for 15 min. The above steps were all tested at RT and thoroughly washed with deionized water after each step. The composition of the plating bath is shown in Table 1. The pH value of the plating bath was adjusted to 8 with ammonia. The temperature of the plating bath was 65 °C and the plating time was 10 min. At last, the product was thoroughly washed with deionized water and dried in a vacuum drying oven at 100 °C.

The SiC coating was then prepared on the surface of the Ni-P coating by the plasma electrolysis spray. The Na_2SiO_3 aqueous solution was used as the electrolyte with the concentration of 50 g/L. The DC voltage between anode and cathode was set as ∼280 V. The detailed process has been previously reported [27–29]. Summarily, the plasma stream and the electrolyte were ejected from the nozzle onto the surface of the carbon fiber, where the SiC coating was rapidly formed on the Ni-P coating. In this case, the laminated nano-coating was formed on the surface of the carbon fiber, with Ni-P and SiC formed as the first and the second layer, respectively.

2.3. Characterization

The surface morphology and composition of the coating were characterized by a scanning electron microscope (SEM) attached with a Links Systems an energy dispersive spectroscope (EDS). The component analysis was characterized by X-ray photoelectron spectroscopy (XPS). In addition, the anti-oxidation property of the samples was measured using a Thermogravimetry (TG) with the heating rate of 20 K/min up to 1000 °C in the air atmosphere.

The electromagnetic parameters (ε', ε'', μ', μ'') of the samples were measured by a vector network analyzer system

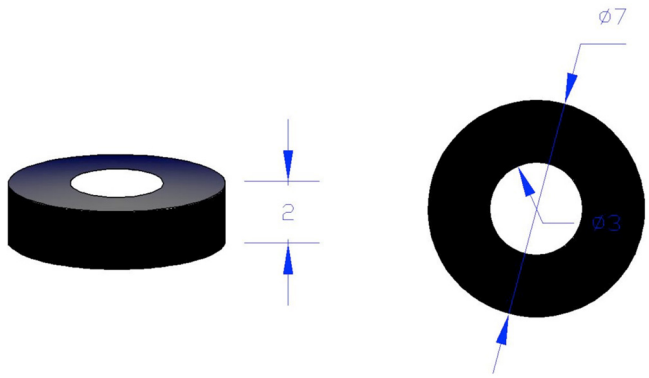

Fig. 2 – The schematic diagram of mixed ring sample (unit:mm).

(VNA, HP-8722ES) within 2–18 GHz at room temperature. The sample for measuring electromagnetic absorption properties was prepared by uniformly mixing the carbon fiber with the paraffin with the mass fraction of the carbon fiber at 3%. The mixture was then filled into a stainless steel mold to form a ring-like sample with the diameter of 7 mm, an inner diameter of 3 mm and a thickness of 2 mm. The schematic diagram of mixed ring samples is shown in Fig. 2. According to transmission line theory, the reflection loss (RL) values of samples were calculated and thickness from 1.0 to 3.0 mm by the following equations:

$$RL = 20\log\left|\frac{Z_{in} - Z_0}{Z_{in} + Z_0}\right| \quad (1)$$

$$Z_{in} = Z_0\left(\frac{\mu_r}{\varepsilon_r}\right)^{1/2}\tanh\left\{j\left(\frac{2\pi fd}{c}\right)(\mu_r\varepsilon_r)^{1/2}\right\} \quad (2)$$

$$Z_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}} \quad (3)$$

where Z_{in} is the input impedance of the absorber; Z_0 is the impedance of free space, f is the frequency of the electromagnetic wave, d is the thickness of the absorber, and c is the velocity of light, μ_0 and ε_0 are the complex permeability and complex permittivity in vacuum, respectively, μ_r and ε_r are the relative complex permeability and the relative complex permittivity, respectively.

3. Results

3.1. *Microstructure of coating*

Fig. 3 shows the SEM images and EDS analysis of the coating. The bare sample was also tested for comparison purpose. Fig. 3a shows the surface morphology of the bare carbon fiber. The sample shows a typical longitudinal ridges and grooves along the fiber axis. Fig. 3b shows the surface morphology of the Ni-P coated carbon fiber. It can be seen that a large number of particles are formed on the surface, but there is no continuous coating formed. Fig. 3c shows the surface morphology of the Ni-P/SiC coated sample. The grooves almost disappeared. The SiC particles fills gaps in Ni coating and forms a uniform coating. The coating was uniform and dense, and also wrapped with carbon fiber. Many SiC nano-dots were uniformly distributed on the Ni-P electroless coatings. The Ni-P coating not only protected the carbon fiber, but also had good extension and thermal conductivity. The SiC coating was deposited quickly and uniformly under the action of plasma. The coating was well bonded to the carbon fiber, and there were no microcracks or micropores. Meanwhile, there was no spallation at the coating. Fig. 3(c1) showed the cross-sectional topography of the coating. The thickness of the coating was measured to be 392.8 nm. The deposited coating covers the carbon fiber surface uniformly and completely. The content of Ni and P was detected to 66.92 wt.% and 22.65 at.%, respectively. The content of Si in the outmost SiC coating was 1.06 wt.% (Fig. 3d).

3.2. *XPS analysis*

The elemental components analysis of the laminated nano-coating was performed by XPS in Fig. 4. As shown in Fig. 4a, the characteristic spectra of Si2p (102 eV), P2p (132 eV), C1 s (283.4 eV), O1 s (528 eV), $Ni2p_{3/2}$ (855 eV)and $Ni2p_{1/2}$ (873 eV). Two Si2p peaks at 102.85 eV (Si-O) and 101.81 eV (Si-C) were separated from the Si2p XPS spectra (Fig. 4b). The Si-O relative content was 16.1% compared to 83.8% of the Si-C relative content (Table 2), reflecting that a small amount of SiO_2 was mixed in the SiC coating. Fig. 4d shows two O1 s peaks at 532.04 eV (Si-O) and 530.57 eV (Ni-O). The Si-O relative content was 64.6% and the Ni-O relative content was 35.3%, which indicated the existence of NiO. Obviously a small amount of Ni was oxidized to form NiO during plasma electrolytic spraying.

3.3. *Anti-oxidation behavior*

In order to characterize the oxidation resistance of the coated carbon fiber, TG and DTG were used to test the bare and coated samples. Fig. 5 shows the TG and DTG curves in air atmosphere from room temperature to 1100 °C. The initial oxidation temperature of the bare sample was 600 °C. It was found that the mass of the coating began to increase at about 400 °C due to the oxidation of Ni to NiO ($2Ni + O_2 = 2NiO$). The DTG curves shows the maximum weight gain rate at 579 °C. Then the sample started to lose weight at around 750 °C, and loses weight completely at 1050 °C. Compared with the bare sample, the oxidation rate was significantly lower, which is due to the combined effect of SiO_2 and SiC. The SiC was oxidized to SiO_2 ($SiC+2O_2 = SiO_2 + CO_2$) during heating. Meanwhile, SiO_2 has fluidity and ductility to protect carbon fibers. As can be seen from Fig. 5b, the maximum weight loss rate was achieved at 904 °C. The bare sample has completely lost weight at 950 °C, and the weight loss rate is the highest at 785 °C. The above results again showed the much enhanced oxidation resistance of the coating for the carbon fiber.

3.4. *Electromagnetic wave-absorbing properties*

The electromagnetic parameters (ε',ε'',μ',μ'') of the paraffin composites containing 3 wt% of carbon fiber and the coating as fillers, were measured at room temperature in the frequency range of 2–18 GHz for the investigation of the electromagnetic

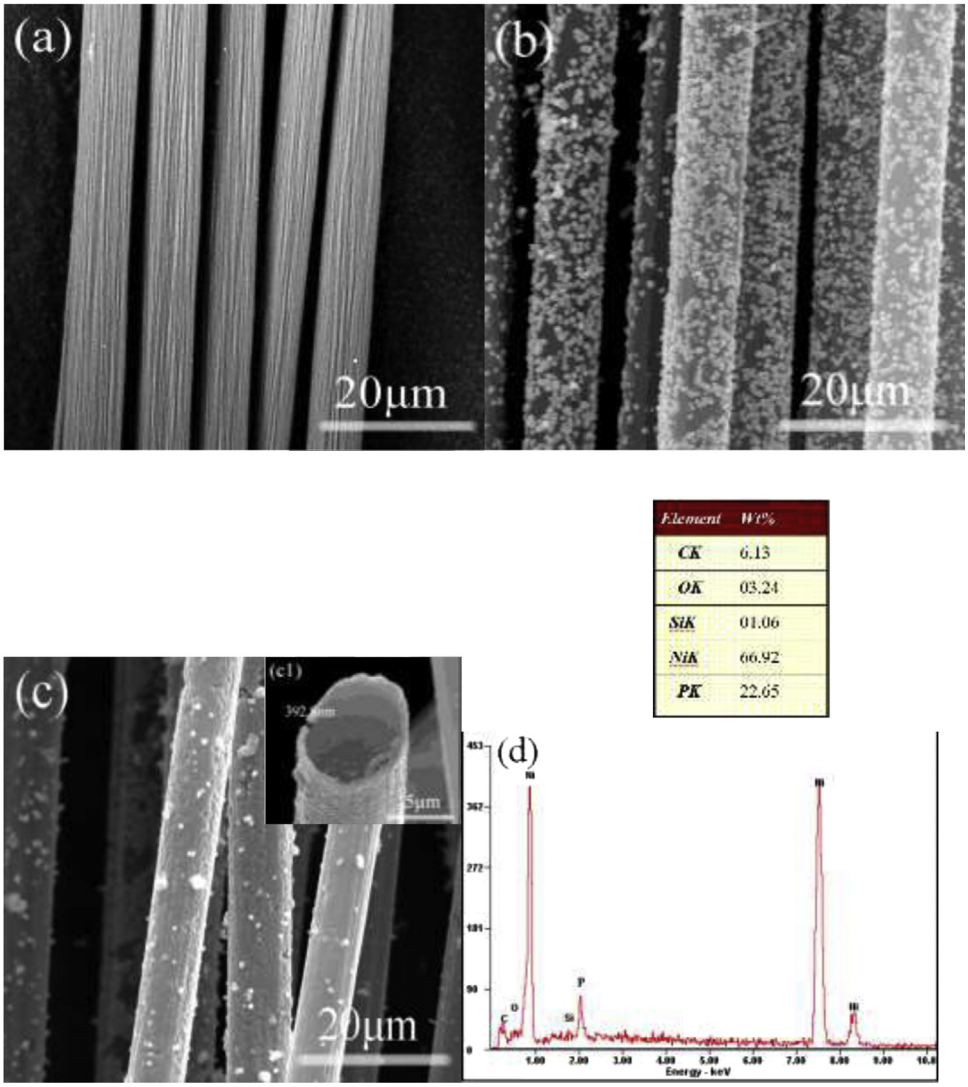

Fig. 3 – Surface morphologis of the bare (a),Ni-P coated fiber (b), Ni-P/SiC coated fiber (c) and the cross-sectional images of the laminated coating on carbon fiber (c1), EDS analysis of Ni-P/SiC coated sample (d).

Table 2 – The binding energy and relative content of O 1s and Si 2p of Fig. 3.

Item	Functional group	Binding energy (ev)	Peak area	FWHM (ev)	Relative content (%)
O	Si-O	532.04	159.67	2.34	64.6%
1 s	Ni-O	530.57	87.34	1.80	35.3%
Si	Si-O	102.85	131.178	0.87	16.1%
2p	Si-C	101.81	681.60	1.39	83.8%

wave absorption properties in Figs. 6 and 7. Relatively complex permittivity ($\varepsilon_r = \varepsilon' - j\varepsilon''$) and relative complex permeability ($\mu_r = \mu' - j\mu''$) indicate the interaction of materials with electric and magnetic fields in electromagnetic radiation, respectively. In order to achieve the maximum absorption of electromagnetic waves, it is necessary to satisfy the impedance matching condition $Z_{in} = Z_0$. According to Eq. 2, if there is a huge difference between the complex permittivity and the magnetic permeability, it is not conducive to impedance matching.

Fig. 6 shows the complex permittivity and complex permeability spectra of the bare and the coated carbon fiber. It can be seen from Fig. 6a and d that ε' for the bare and the coated carbon fiber were in the range of 28.60−14.51 and 19.79−7.88, respectively. Similarly, the values of ε'' were located in the range of 56.75−14.72 and 22.15−8.35, respectively. Compared with the bare carbon fiber, the values of ε' and ε'' of the coated fiber were significantly decreased. The increase in the resistivity and space charge polarization after deposition of the coating were mainly resulted from the decrease in dielectric constant. The same situation was shown in Fig. 6b, where the values of μ' and μ'' of the bare carbon fiber were in the range of 0.81–2.13 and −0.56−0.81, respectively. Meanwhile, as presented in Fig. 6e, the values of μ' and μ'' of the coated fiber were in the range of 0.81–1.91 and −0.51−0.68, respectively. The

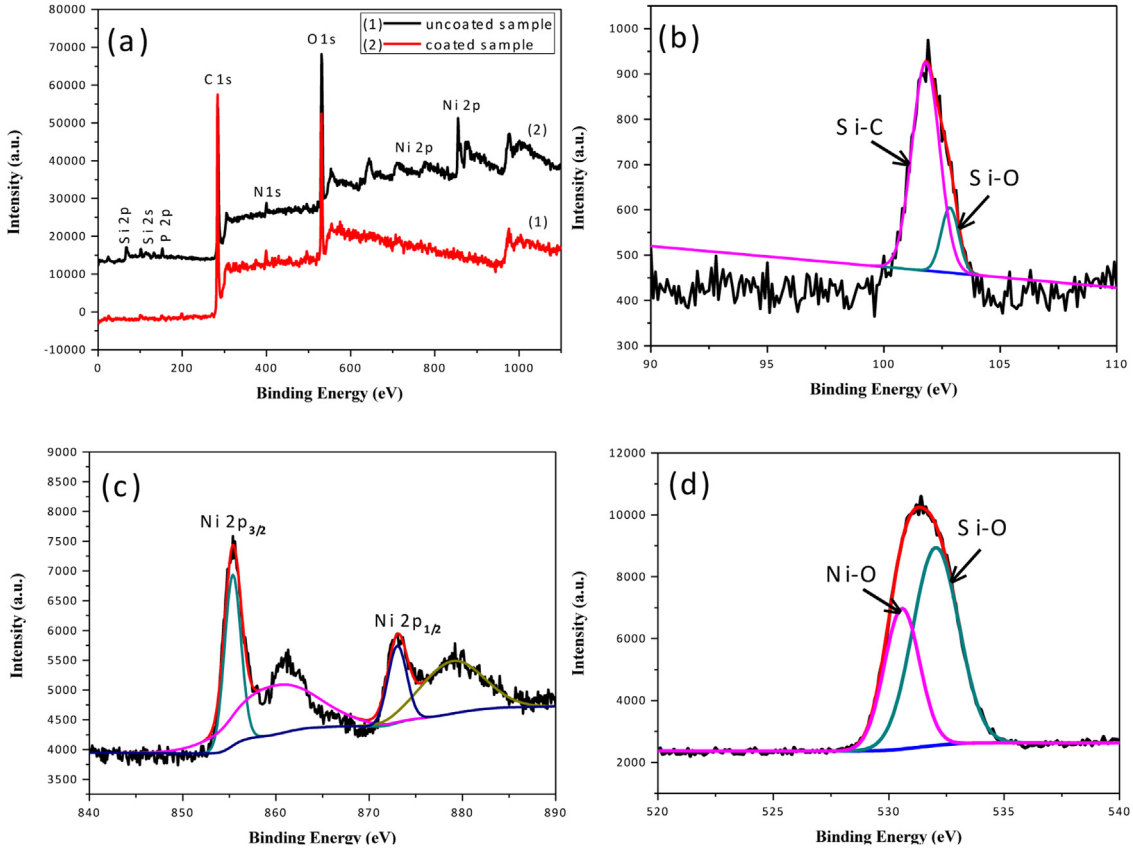

Fig. 4 – XPS survey spectra (a) and high resolution of (b) Si 2p, (c) Ni 2p, (d) O 1s.

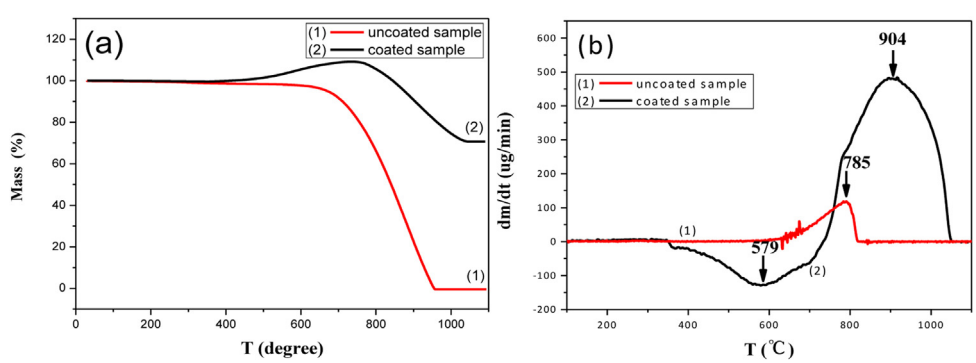

Fig. 5 – TG (a) and DTG (b) curves of the bare and Ni-P/SiC coated samples.

paraffin in the absorbing ring sample is a high loss dielectrics, therefore, the negative μ" may be caused by the resonance of the coated carbon fiber and paraffin mixture [30].

The electromagnetic wave absorption properties of materials are normally caused by their dielectric loss and magnetic loss. In order to study the main loss factors of the coated fiber, the dielectric loss factor ($\tan\delta_\varepsilon = {}^{\varepsilon''}/_{\varepsilon'}$) and the magnetic loss factor ($\tan\delta_\mu = {}^{\mu''}/_{\mu'}$) were calculated from the complex permeability and the complex permittivity, respectively. As shown in Fig. 6c and f, the dielectric loss tangent was larger than the magnetic loss tangent, indicating that the electromagnetic wave absorption performance of the bare and the coated carbon fibers mainly derives from dielectric loss.

Cole-Cole diagram was plotted by the experimental permittivity of bare and coated carbon fiber to further interpret the mechanisms of the permittivity which presented the Debye dielectric relaxation process. It can be analyzed through the Debye relaxation equation:

$$\left(\varepsilon' - \frac{\varepsilon_s + \varepsilon_\infty}{2}\right)^2 + \left(\varepsilon''\right)^2 = \left(\frac{\varepsilon_s - \varepsilon_\infty}{2}\right)^2$$

where ε_s and ε_∞ were the static permittivity and relative dielectric permittivity at the high-frequency limit, respectively. Thus, the plot of ε' versus ε" would be a single semicircle and corresponds to one Debye relaxation process.

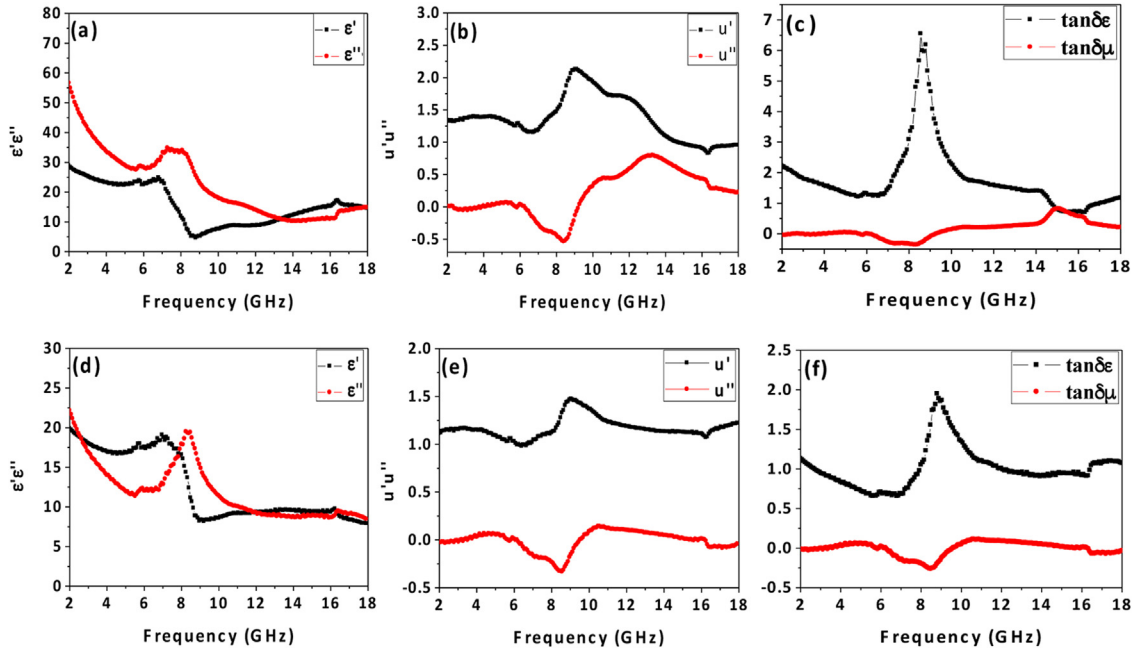

Fig. 6 – Frequency dependence of the complex permittivity (a, d), complex permeability (b, e), dielectric loss factor and magnetic loss factor(c, f) of the bare (a,b,c) and the Ni-P/SiC coated (d,e,f) carbon fibers.

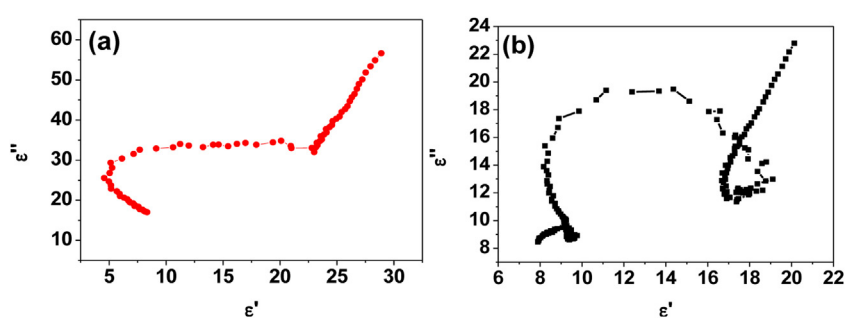

Fig. 7 – Cole–Cole diagram for (a) bare carbon fiber and (b) Ni-P/SiC coated carbon fiber.

As shown in Fig. 7a, there was a semicircle and the semicircle was distorted, indicating that there may be other absorption mechanisms, such as conductivity loss, interfacial polarization, and defects. There were three Cole-Cole semicircles, suggesting that there are ternary Debye dipolar relaxation process (Fig. 7b). The debye relaxation process can enhance the dielectric properties of composites. Each semicircle means a Debye relaxation process related to the interfacial polarization effect, which greatly helps the dielectric loss.

Fig. 8a and b show the reflection loss for the bare and the coated carbon fibers at different thicknesses. Generally, the RL value was below 10 dB, and more than 90% of the electromagnetic wave energy was absorbed. The corresponding frequency range was usually defined as the effective absorption bandwidth, and the absorbing material was considered to be a qualified electromagnetic wave absorber. As shown in Fig. 8a, the minimum reflectivity of the bare carbon fiber reached -3.9 dB at 10.17 GHz when the thickness is 1.5 mm. After the laminated coating was formed on carbon fiber, the RL values were significantly enhanced. The minimum RL value of −26.95 dB was achieved at 11.35 GHz with the thickness of 2.8 mm, as shown in Fig. 8b. It was revealed that 99% of the electromagnetic waves were absorbed. Meanwhile, the sample exhibited a double-peak absorption characteristic. When the thickness was 2.5 mm, the minimum RL value reached −16.43 dB at 15.22 GHz. It can be seen from the above results that after the carbon fiber was coated with the laminated coating, the RL value was below 10 dB in the ranges of 10.17–12.31 GHz, 14.26–16.08 GHz band, reflecting that the sample exhibited good electromagnetic wave absorption performance.

4. Discussion

Based on the above experimental results, it could be clearly revealed that the laminated coating effectively improved the electromagnetic wave absorption performance of the carbon fiber. The Ni-P electroless coating was prepared on the surface of the carbon fiber, and then the SiC nano-coating was formed on the surface of the Ni-P coating by electrolytic plasma spray-

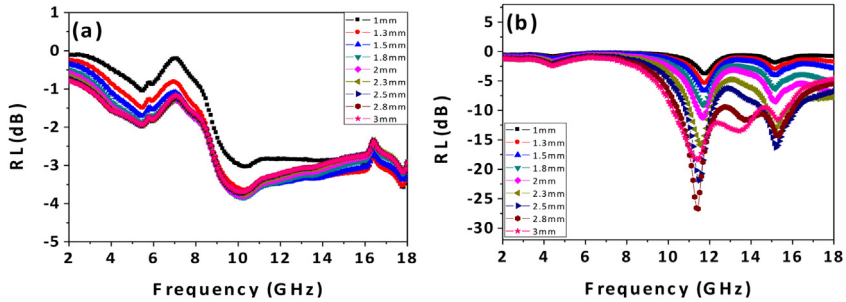

Fig. 8 – The reflection loss for the bare (a) and the Ni-P/SiC coated (b) carbon fiber at different thicknesses.

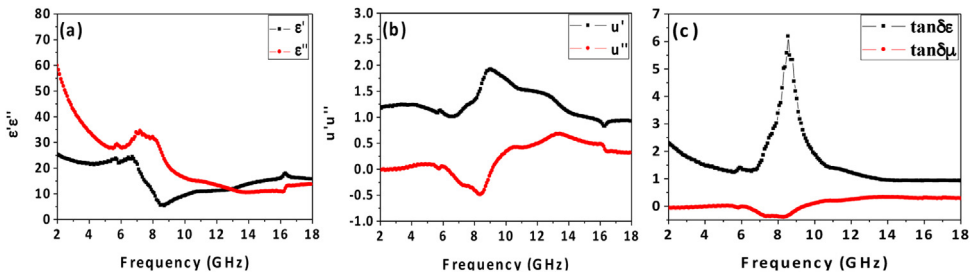

Fig. 9 – Frequency dependence of the complex permittivity (a), complex permeability (b), dielectric loss factor and magnetic loss factor (c) of the Ni-P-coated carbon fiber.

ing. The Ni-P coating not only protected the carbon fiber from plasma damage, but also made the ceramic coating easy for deposition. Meanwhile, the laminated coating improved the high-temperature oxidation resistance and absorbing properties of the carbon fiber. SiC is a typical high temperature dielectric loss absorbing material. There were multiple heterogeneous interfaces among SiC, Ni-P and the carbon fiber, which introduced interfacial polarization and enhanced the electromagnetic waves attenuation capacity.

Fig. 9 shows the complex permittivity, complex permeability spectra, dielectric loss tangent and magnetic loss tangent of the Ni-P electroless coating on carbon fiber. It could be seen that ε' and ε" were in the range of 25.23–4.84 and 60.12–10.29, respectively. The values of μ' and μ"were in the range of 0.80–1.93 and −0.48–0.67, respectively. The complex dielectric constant of Ni-P coating was larger than that of the present coating. Therefore, the SiC coating played a key role in absorbing electromagnetic wave for carbon fiber. As shown in Fig. 9c, the dielectric loss tangent was larger than the magnetic loss tangent, indicating that the electromagnetic wave absorption performance of the bare and the coated carbon fibers mainly derives from dielectric loss. Fig. 10 shows the reflection loss for the Ni-P-coated fiber at different thicknesses. The minimum reflectivity of Ni-P-coated fiber was −7.0 dB at 14.66 GHz when the thickness was 1 mm. Compared with the bare carbon fiber, the minimum RL moved toward the high frequency region, but a large number of electromagnetic waves were still reflected. The minimum RL of the laminated coating is −26.95, indicating that the SiC coating made a great contributes to the absorption electromagnetic waves.

The electromagnetic wave attenuation mechanism is first to make electromagnetic waves enter the material, and then

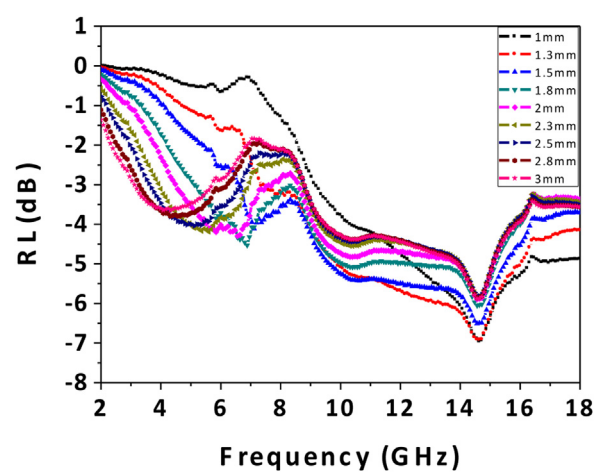

Fig. 10 – The reflection loss of the Ni-P-coated carbon fiber at different thicknesses.

convert the incoming electromagnetic waves into thermal energy or other forms of energy through the material itself. But to make all the electromagnetic wave to enter the material, it is necessary to match the material with free space as much as possible. Ideally, when $Z_0 = Z_{in}$, the electromagnetic waves do not reflect and all enter the material interior.

The electromagnetic wave reflection coefficient is defined as:

$$R = \frac{Z_{in} - Z_0}{Z_{in} + Z_0} \tag{4}$$

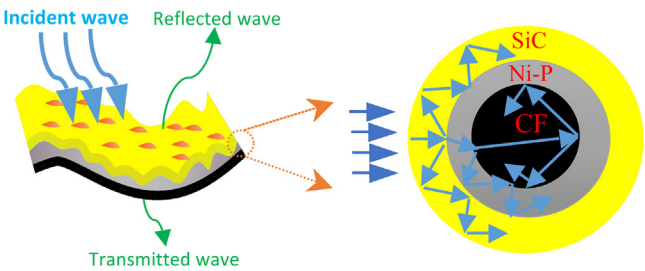

Fig. 11 – The schematic diagram of the electromagnetic wave absorption mechanism of the laminated coating on carbon fiber.

The absorbing material impedance can be expressed as:

$$Z_{in} = \sqrt{\frac{u_0 u_r}{\varepsilon_0 \varepsilon_r}} \quad (5)$$

In order to satisfy the material reflectivity R = 0, that is, only $\varepsilon_0 = \varepsilon_r$ can achieve zero reflection of electromagnetic waves. The absorbing materials which can satisfy this condition have not been found in practice. However, a single absorbing material and a single layer material are difficult to meet the theoretical requirements. Therefore, multi-layer or composite absorbents are generally designed to achieve better matching and maximum absorptive capacity.

According to the above analysis, a feasible electromagnetic wave absorption model diagram is presented in Fig. 11. When incident electromagnetic waves were incident on the surface of the coating by air, only a small amount of electromagnetic waves were directly reflected back into the air by the surface of the coating due to the excellent impedance matching characteristics. Another part of the electromagnetic wave was transmitted into the interior of the absorbing coating. The electromagnetic wave was further dissipated by the synergistic effect of the laminated coating. At the same time, the electromagnetic waves entering the material undergo multiple reflections between the interfaces. Interfacial polarization effects also play an important role in dielectric loss. All these factors are the causes of electromagnetic wave loss. Therefore, only a small amount of electromagnetic waves can be reflected and/or passed through the laminated coating.

5. Conclusion

A novel metal/ceramic laminated nano-coating (Ni-P/SiC) was prepared on the surface of carbon fiber, that is, the Ni-P coating was initially electroless plated on the carbon fiber, and then SiC coating was synthesized on the Ni-P by the electrolytic plasma spray. Compared with the pure Ni-P coating, the electromagnetic wave absorption performance of the composite material was greatly improved and exhibits a bimodal absorption characteristic. The Ni-P/SiC coating demonstrated a double-peaks absorption characteristic. The minimum RL value of −26.95 dB was observed at 11.35 GHz for the thickness of 2.8 mm, while the RL values of −16.43 dB was achieved at 15.22 GHz. The superior absorption properties of the material were attributed to the synergistic effect of dielectric loss and good impedance matching. The oxidation resistance of carbon fiber was effectively improved with the maximum weight loss rate increased from 785 to 904 °C.

Conflict of interest

The authors declare no conflicts of interest.

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